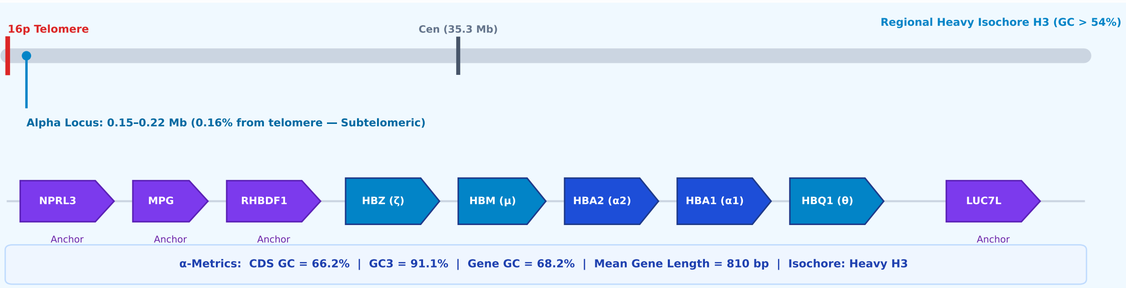
Figure 0: Conceptual Paradigm, Syntenic Discovery Pipeline, Metric Anatomy & Interactive Dashboard User Guide

A Macroevolutionary Atlas of Vertebrate Globin Clusters across 68 Genomes • Resolving Isochore Polarity, Syntenic Translocations, and Codon Biophysics

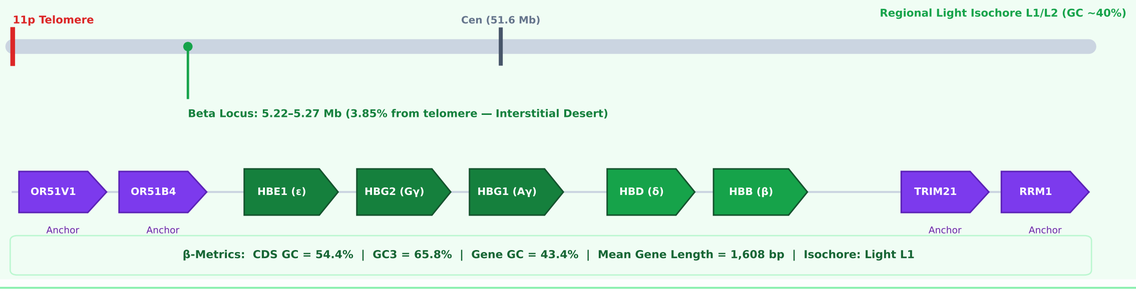
A. The Human Globin Paradigm & The Core Evolutionary Inquiry

Canonical dual-cluster arrangement on chromosomes 16 and 11, and the hypothesis of universal isochore polarity

Human Alpha-Globin Cluster — Subtelomeric Heavy Isochore (16p13.3)



Human Beta-Globin Cluster — Interstitial Light Isochore (11p15.4)



The Core Macroevolutionary Inquiry:

1. Is this extreme GC disparity (subtelomeric high-GCα vs. interstitial low-GC β) an invariant vertebrate feature?
2. Did physical distance to the telomere dictate locus GC content via GC-biased gene conversion (gBGC) across 500 Myr?

B. Synteny-Anchored Locus Retrieval Pipeline Across 68 Genomes

Identification of homologous intervals via conserved syntenic anchors and discovery of 4 macroevolutionary archetypes

Synteny Discovery Pipeline & Chromosome Contiguity Filtering

1. Syntenic Search

Query 68 genomes for flanking markers:
α: NPRL3, MPG, RHBDF1, LUC7L
β: OR51 cluster, CDCA8, TRIM21, RRM1

2. Interval Bounds

Define syntenic window containing all flankers and globin duplications.
Extract verified coding models.

3. Scaffold vs Chr

Discriminate assembled chromosomes from unplaced scaffolds to avoid artificial telomeric misassignment.

4. Baseline Calibration

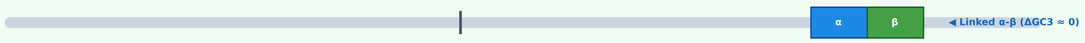
Compute whole-genome CDS & gene baselines (mean, SD, percentiles) to calculate locus z-score deviations.

1. Linked Primordial Archetype (Basal Actinopterygii & Amphibians — Spotted Gar, Xenopus)

Ancestral tandem arrangement: α and β globin genes reside on the same chromosome in a single cluster.

Both genes share identical genomic environment and regional recombination rate. ΔGC3 (α - β) ≈ 0%.

Cytogenetic State: Single Chromosome (Gar Chr 19 / Xenopus Chr 1)



2. Canonical Decoupled Subtelomeric-α (Eutherian & Metatherian Mammals — Human, Mouse)

Amniote translocation decouples α and β. α captured by subtelomere (<1 Mb to telomere); β remains interstitial.

Runaway subtelomeric gBGC supercharges α-GC3 (>85–95%) while β-GC3 remains moderate (~60–65%). ΔGC3 >> 0%.

Cytogenetic State: Two Chromosomes (α Subtelomeric / β Interstitial)

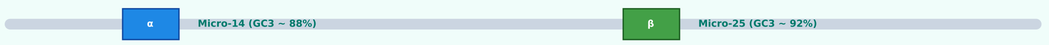


3. Dual Microchromosomal Partitioning (Archosauria & Aves — Chicken, Eagle, Hummingbird)

Both α and β clusters translocate onto distinct, highly recombinogenic microchromosomes (Chr 14 and Chr 25).

High crossover density drives synchronous hyper-elevation of both clusters: α-GC3 > 85%, β-GC3 > 85%. ΔGC3 ≈ 0%.

Cytogenetic State: Two Microchromosomes (Micro-14 and Micro-25)



4. The Great Inversion (Squamate Reptiles — Lizards & Snakes — Wall Lizard, Garter Snake)

Karyotypic inversion translocates β-locus to microchromosome/subtelomere, leaving α in an interstitial desert.

Inverted gBGC polarity: β-GC3 skyrockets (>90–96%) while α-GC3 languishes (~48–55%). ΔGC3 << 0% (β >> α)!

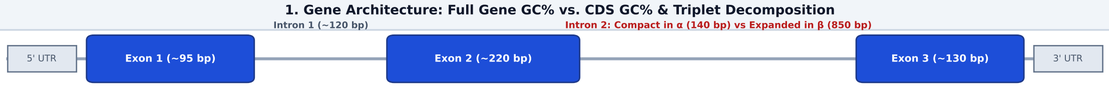
Cytogenetic State: Inverted Polarity (β Subtelomeric / α Interstitial)



C. Anatomy of Calculated Metrics: Gene Architecture & Codon Physics

Biophysical basis of CDS GC%, GC3%, Whole-Gene GC%, Intron Compaction, and Direct Synonymous Shifts

1. Gene Architecture: Full Gene GC% vs. CDS GC% & Triplet Decomposition

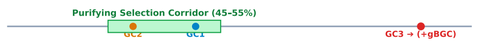


Whole-Gene GC%: Exons + Introns + UTRs (Reflects regional chromatin & isochore baseline GC)

CDS GC%: Coding sequence only (Concatenated Exons 1, 2, 3; reflects amino acid & mutational bias)

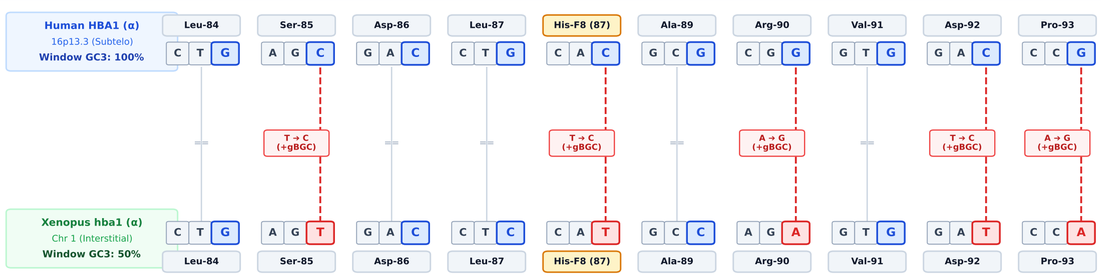
Codon Triplet Breakdown:

- GC1: Moderate purifying constraint (~50-65% GC)
- GC2: Locked in Purifying Corridor (45-55%)
- GC3: Synonymous site! Barometer of gBGC (35-98% GC)
- GC4d: 4-Fold degenerate (100% synonymous; pure gBGC)



2. Conserved Globin Heme Pocket: Proximal Histidine (His-F8) Motif & Synonymous GC3 Divergence

Comparison across 350+ Myr: 100% Identical Amino Acids, Yet 100% of Divergent 3rd Positions Shifted to G/C in Human!



3. Bernardi's Compaction Law & Telomeric Distance Driver

Bernardi's Compaction Law: High-GC isochores suppress intron expansion; low-GC accumulate massive introns.

Human HBA1 (α): 843 bp (Intron 2 = 140 bp) [Compact High-GC] vs. Human HBB (β): 1,608 bp (Intron 2 = 850 bp) [Expanded Low-GC]

Distance-to-Telomere (dtelo / Lchr) Drives gBGC:

- Meiotic crossovers are physically biased toward chromosome ends, accelerating GC-biased gene conversion (gBGC).
- **Subtelomeric loci (dtelo < 1 Mb) undergo runaway GC3 elevation; internal/centromeric loci remain buffered in AT-rich baselines.**

`Relative Telomeric Distance: dtelo_rel = min(Pos, Chr_Len - Pos) / Chr_Len  (0.000 Subtelomeric → 0.500 Metacentric)`

D. Interactive Dashboard Architecture & Analytical User Guide

How to navigate and interpret the 5 interconnected exploratory modules in dashboard.html

Tab 1: Global Macroevolutionary Figures `[ Fig 1A ]` `[ Fig 1B ]` `[ Fig 2 ]` `[ Fig 3 ]`

The Big-Picture Evolutionary Synthesis

- Fig 1A: Curated benchmark summary across 33 representative species.
- Fig 1B: Exhaustive distribution across all 68 vertebrate genomes.
- Fig 2: Mechanistic drivers (Bivariate GC regressions, gene length compaction, telomere distance).
- Fig 3: Time-calibrated phylogenetic chronogram, chromosome synteny maps, and codon ΔGC3 divergence.

Tab 2: 1-on-1 Head-to-Head Comparison `[ Linear Tracks ]` `[ Inset Map ]` `[ Diagnostic Suite ]`

Direct Pairwise Syntenic & Structural Contrast

- Select Species A & B from searchable comboboxes (e.g., Human vs. Spotted Gar).
- Synchronized views: Linear Chromosome tracks, Bivariate with Mini Inset, or 6-Panel Diagnostic suites.
- Instant differential readout: ΔCDS GC%, ΔWhole-Gene GC%, and chromosomal context.

Tab 3: Single Species Deep Dive `[ Interval Bounds ]` `[ Flanking Markers ]` `[ Ideograms ]`

Locus Anatomy & Architectural Metadata

- Comprehensive profile of any individual species in the 68-genome database.
- Exact genomic interval bounds, flanking syntenic markers, and gene count.
- High-resolution chromosome ideograms displaying exact physical coordinates.

Tab 4 & 5: Interval Database & Statistical Contrasts `[ 1,530 Gene Models ]` `[ Wilcoxon Tests ]` `[ Effect Sizes ]`

Curated Gene Models & Hypothesis Testing

- Tab 4 (Database): Filterable, searchable table of all 1,530 curated gene models with z-scores.
- Tab 5 (Statistics): Two-sided Wilcoxon rank-sum tests, Hodges-Lehmann effect sizes, and FDR q-values.